

Abdullah Al-Nafisah

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MSc, Electrical and Computer Engineering

2022 – 2024

King Abdullah University of Science and Technology (KAUST), Thuwal, KSA | GPA 3.83/4.0

Thesis: *Turbo-Equalized FTN-Inspired Modulations*, doi:10.25781/KAUST-M5140

BSc, Double Major in Electrical Engineering & Physics

2013 – 2019

King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, KSA | GPA 3.76/4.0, 1st Honors

Capstone: *Prototype of a Wireless Seismic Data Acquisition System*, Best Senior Project Award, 2019

Exchange Student

Fall 2016

Georgia Institute of Technology, Atlanta, GA, USA | GPA 3.5/4.0

Summer School on Superconducting Electronics

Summer 2024

FLUXONICS, The European Foundry for Superconducting Electronics e.V., Corsica, France

Research & Development

Oct 2020 – Present

National Company of Telecommunications and Information Security (NTIS), KSA

- **Qubit control on the RFSoc:** RF system on the AMD/Xilinx RFSoc ZCU216 with Pulse and CW modes and native LOs, VNAs, digitizers, and AWGs; 2 dBm output, worst spur 35 dBc below the carrier from 1.5 to 7.5 GHz.
- **Real-time classification in fabric:** FPGA-based flexible line qubit classifier for mid-circuit measurement and active reset at about 50 ns. A trigger mode from a TTL-compatible sync signal mitigates vibration-induced errors.
- **Qubit characterization:** Single-qubit calibration (resonator punch-out, two-tone spectroscopy, Rabi, Ramsey) with a trained single-shot readout discriminator; resonators measured at high and low power for single-photon drive power.
- **Quantum links:** QKD transmitter and receiver on the ZCU102, GTH transceivers sending on-off keyed pulses at 1 Gbps through an intensity modulator and attenuators, down to single photons at the detector after 50 km of fiber. FPGA entropy extraction for QRNG on the IDQ20MC1 chip, certified to NIST SP 800-90B.
- **RF and embedded:** Direct RF sampling with x16 DACs and ADCs on RFSoc, optical links on MPSoC, off-the-shelf RF switches for multiplexing, PCB design, and VHDL/Verilog controllers with custom UART, I2C, and SPI drivers.

Avionics Engineer

Dec 2019 – Oct 2020

PSATRI, Riyadh, KSA

- PCB design for unmanned aerial vehicle avionics in Altium, with weight, thrust-to-weight ratio, and battery sizing optimized for about 30 minutes of flight time.

Master's Thesis Research

Spring 2023 – Spring 2024

Supervised by Dr. T. Al-Naffouri and Dr. M. Siala at KAUST, Thuwal, KSA

- Integrated FTN-inspired modulations with turbo equalization, convolutional and LDPC decoding, beating turbo-equalized 4-ASK by 2.3 dB at BER 10^{-5} , at the same two bits per channel use.

Guided Research on Analog Electronic Circuits

Fall 2018 – Spring 2019

Supervised by Dr. M. Abuelma'atti at KFUPM, Dhahran, KSA

- Built a memristor-based chaotic masking system from off-the-shelf components, validated synchronization of two chaotic oscillators, and designed a tunable CFOA comb filter.

Mitacs Globalink Research Internship

Summer 2018

Supervised by Dr. S. Valluri at Western University, London ON, Canada

- Applied the Lambert W function to the I-V characteristics of diode-based circuit models, supporting work the group carried on for several years.

Visiting Student Research Program (VSRP) Internship

Summer 2017

Supervised by Dr. H. Fariborzi at KAUST, Thuwal, KSA

- Modeled All Spin Logic circuits in Cadence with Verilog-A, applying the Landau-Lifshitz-Gilbert equations and compact charge/spin conduction modules.

Projects

github.com/AbdullahAlNafisah

- **llm-kv260-demo:** An LLM decoded by four AMD Kria KV260 FPGAs on one stream.
- **coded-msprs:** Rate-2 turbo-equalized MS-PRS chain in C++ on AFF3CT, behind the thesis and the NSM paper.
- **cube-gradiometer-esp32:** DIY magnetic gradiometer, four BMM350 magnetometers on a 3 cm cube behind an I2C mux, opposite pairs differenced for two columns of the gradient tensor; ESP32 firmware, MATLAB analysis.

Papers

1. M. Siala, A. Al-Nafisah, and T. Al-Naffouri, "Nyquist Signaling Modulation (NSM): An FTN-Inspired Paradigm Shift in Modulation Design for 6G and Beyond," *arXiv:2511.08553*, 2025.
2. A. Al-Nafisah, D. Khan, S. Amara, and Y. Massoud, "Synchronization of Two Transistor-Based Chaotic Oscillators: Experimental Verification," *IEEE 65th International Midwest Symposium on Circuits and Systems (MWSCAS)*, Fukuoka, Japan, pp. 1–4, 2022.
3. M. T. Abuelma'atti and A. Al-Nafisah, "New CFOA-Based Shadow Analog Comb Filter," *Journal of Active and Passive Electronic Devices (JAPED)*, vol. 15, no. 1–2, pp. 1–8, 2020.
4. M. T. Abuelma'atti and A. Al-Nafisah, "Synchronization of Memristor-Based Chaotic Oscillator: Experimental Verification," *Journal of Active and Passive Electronic Devices (JAPED)*, vol. 15, no. 1–2, pp. 21–28, 2020.
5. M. T. Abuelma'atti and A. Al-Nafisah, "A Memristor-Based Chaotic-Masking for Analog Spread-Spectrum Communication," *Indonesian Journal of Electrical Engineering and Computer Science (IJECS)*, vol. 14, no. 2, pp. 966–971, 2019.